Testing Physics-Based Rules for Accurate SEM-EDS of Magnesium in Additively Manufactured AlSi10Mg

Applicant: Ronnie Guymon, junior in Mechanical Engineering, Brigham Young University
Supervisor: Prof. Sterling G. Baird, BYU · **Period:** January to December 2027 · **Request:** $3,000
Facility: BYU Electron Microscopy Facility (Thermo Fisher Apreo SEM, EDAX Octane Plus EDS)

1. Introduction

Quantitative energy-dispersive X-ray spectroscopy (EDS) in a scanning electron microscope (SEM) is only as good as the settings used to collect the spectrum. Beam energy, beam current, and acquisition time control how many X-rays the sample produces, how many are absorbed before they leave it, and how many artifacts the detector adds. With measured standards, EDS can reach about 5% relative uncertainty. Standardless analysis can be off by 30%, and normalizing the total to 100% hides the error [1]. Most operators use general defaults, such as 15 to 20 kV and "enough" counts, that were not designed for light, low-concentration elements in new alloys.

Our group is building CALIBER, a software tool that recommends SEM, EDS, EBSD, and XRF settings for a given sample and measurement goal. It combines the published literature, physics models, and our own measurements, and it updates its recommendations as new data come in. Its first test material is AlSi10Mg made by laser powder bed fusion (LPBF). Magnesium makes up less than 1 wt% of this alloy, yet it controls precipitation hardening, and its X-ray peak sits on the shoulder of an aluminum peak more than 100 times stronger. Our literature review found no controlled study of SEM-EDS settings for Mg in LPBF AlSi10Mg that was checked against an independent bulk analysis. The settings in published work are examples, not tested choices.

My first measurements show that the problem is real (Table 1). The vendor's standardless software reported 1.64 wt% Mg at 15 kV and 0.94 wt% at 5 kV, both well above the 0.20 to 0.45 wt% that the powder supplier specifies. At 5 kV the Mg peak is clearly detected, about 10σ above background, but its net intensity changes by about 30% depending on how the aluminum tail and background beneath it are modeled, three times its counting error. When I ran the vendor's k-ratios through the independent matrix correction in NIST DTSA-II [2], Mg still came out at 0.90 wt%. The error is in the measured peak intensities, not the correction, so more counts alone will not fix it. The 15 kV result also failed a physics check: from 5 to 15 kV the Mg-to-Al intensity ratio rose by a factor of 1.52, where X-ray generation and absorption predict a drop to 0.86. Since September I have measured the probe current with a Faraday cup and collected 5 kV spectra from 0.77 to 5.6 million counts, including eight replicates of 1 million counts each. In November, on a newly polished sample, the C and O peaks were clear at 5 kV, only a small O peak remained at 10 kV, and both were nearly gone at 20 kV, as expected for a thin surface film.

Table 1. Preliminary EDS of one LPBF AlSi10Mg sample (vendor standardless quantification, normalized to 100%).

Session	Beam energy	Total counts	Live time	Mg (wt%)	Si (wt%)	O (wt%)
April 30, 2026	15 kV	4.44 million	461 s	1.64	9.47	8.10
August 2026	5 kV	0.65 million	328 s	0.94	11.15	1.68
Powder specification				0.20–0.45	9.0–11.0	

To explain these results, I proposed a rule for choosing beam energy, which we then refined together. X-rays made deep in the sample leave along a slanted path to the detector that is 1.74 times their depth at our 35.1° take-off angle, and some are absorbed on the way out. The rule says a line can be quantified reliably only if this exit path, measured from the deepest point where the line is produced [3], is shorter than one attenuation length for that X-ray. For AlSi10Mg, Si Kα sets the limit at 7.6 kV. Of the 5, 10, 15, and 20 kV settings in routine use, only 5 kV should give clean results for Al, Si, and Mg. O Kα fails at all

four, in line with long-standing advice not to quantify lines below 1 keV [4]. Related rules for beam current and acquisition time follow from dead time and counting statistics [5, 6]. This project tests these rules against an independent measurement of the true composition and turns the results into CALIBER's first validated recommendations.

2. Goals

1. **Measure the true composition.** Measure Mg in the printed sample and in its starting powder by ICP-MS, so EDS accuracy is judged against a measurement instead of a nominal value. Comparing the two also shows whether Mg evaporated during printing.
2. **Test the voltage rule.** Collect standards-based, un-normalized EDS at 5, 10, 15, and 20 kV and choose among three outcomes written down in advance: the rule holds, the rule is too strict, or the rule needs a correction for surface oxide and contamination, which matter most at 5 kV.
3. **Find the usable beam current.** Measure the actual probe current, dead time, and pulse pile-up at each current setting, and choose the dead-time range from data. This settles the gap between the 10% or less that NIST recommends for trace work [6] and the 25 to 30% often used for speed.
4. **Find when longer acquisition stops helping.** Find the count level where the scatter between replicate spectra grows larger than their counting error, so more time no longer improves accuracy, while tracking carbon buildup.
5. **Validate and share.** Enter each tested rule into CALIBER with its measured uncertainty, release the spectra, scripts, and session records openly, and present the results at M&M 2027.

3. Methods

Samples and reference composition. I will section, mount, and polish LPBF AlSi10Mg printed from virgin powder on the University of Utah's Aconity MIDI printer (Prof. Ashley Spear's group). I will record the final polishing step, because embedded colloidal silica would read as Si and O in the top 0.4 µm that the beam samples at 5 kV, and I will store polished samples in a vacuum desiccator and plasma-clean them before each session. For the reference composition, I will work with Prof. Devin Rappleye's group (BYU Chemical Engineering) to digest offcuts of the printed sample and the powder and analyze them by ICP-MS (Agilent 8900). Acid-only digestion does not fully recover Si, so Si will come from wavelength-dispersive XRF (Rigaku Primus II, BYU Geological Sciences) or a digestion that includes hydrofluoric acid.

Instrument, standards, and quality checks. All EDS work uses the Apreo SEM and EDAX Octane Plus silicon-drift detector at the 10 mm analytical working distance. The lab has already bought pure standards: 99.99% Al, a Si wafer, and an MgO crystal, which we section and polish without water because it would not cleave. I will give the standards the same thin carbon coat as the sample and collect them in the same session at every voltage. The pure Al spectrum also gives the real shape of the Al tail under the Mg peak. Each session starts with quality checks: energy calibration, detector resolution, counts per nA·s on the Si wafer, and a Duane–Hunt check for charging [5]. I will read the probe current on the Faraday cup before and after every spectrum and record it with the spectrum, because the k-ratio depends on electron dose.

Experiment 1: voltage series (Goal 2). I will analyze the same marked region at 5, 10, 15, and 20 kV, keeping the amplifier time fixed and adjusting the probe current to hold dead time constant. Replicate scans on nearby areas will add up to about 3% counting precision on Mg, or about 7 million counts at 5 kV. We wrote down seven predictions before collecting data. For example, standards-based Si and Mg should hold steady within counting error at 5 kV and drift in the order the rule predicts at higher voltages. Apparent C and O should fall about tenfold from 5 to 20 kV as a surface film becomes a smaller part of the analyzed volume.

Experiment 2: beam current and dead time (Goal 3). At 5 kV I will step through the current settings and record the measured current, dead time, count rate, the Al sum peak at 2.97 keV, and the Mg-to-Al intensity ratio. The usable dead-time range ends where that ratio moves away from its low-dead-time value by more than the counting error.

Experiment 3: acquisition time (Goal 4). Building on the September replicates, I will collect five or more spectra at each count level on fresh nearby areas and compare their scatter with the counting error each one predicts. I will also compare one long acquisition with several short ones of the same total live time. Tracking how the C peak grows, and whether Mg drifts, as the dose builds will set a dose limit for each spot.

Analysis. I will fit spectra with our Poisson-weighted peak model, which gives net intensities and Currie detection limits [7], and quantify them against same-session standards in DTSA-II with a φ(ρz) matrix correction [2]. I will report un-normalized totals, with 98 to 102% as a health check, next to the vendor's standardless results, and split each condition's error against the ICP-MS value into counting and systematic parts. I will judge each result against the rules written in advance and enter the tested rule into CALIBER with its measured uncertainty, completing the first full pass through CALIBER's feedback loop.

4. Timeline, outcomes, and significance

2027	Work
Jan–Feb	Mount and coat standards; ICP-MS reference composition; quality checks; M&M 2027 abstract
Mar–Apr	Experiment 1: voltage series (four sessions plus standards)
May–Jun	Experiment 2: beam current, dead time, and pile-up
Jul	Experiment 3: acquisition time; enter results in CALIBER
Aug	Present at M&M 2027
Sep–Dec	Journal manuscript, open data release, and extension to the lab's next alloys

Expected outcomes. This project will produce (1) the first SEM-EDS settings study for LPBF AlSi10Mg checked against an independent composition measurement; (2) tested or corrected rules for beam energy, current, and acquisition time, each with a measured uncertainty; (3) an open dataset and analysis scripts that any EDS user can rerun; and (4) CALIBER's first experimentally validated recommendations, which our lab will extend to other alloys. I will prepare the samples, run the sessions, and analyze the data. This will train me in standards-based quantitative microanalysis, which I plan to use in a career developing and analyzing materials for aerospace applications.

References

1. Tong, V. & Mingard, K. *Measurement uncertainties of energy dispersive X-ray spectroscopy in the SEM (SEM-EDX/EDS)*. NPL Report MAT 135 (2026). doi:10.47120/npl.mat135
2. Ritchie, N. W. M. Spectrum simulation in DTSA-II. *Microsc. Microanal.* 15, 454–468 (2009). doi:10.1017/S1431927609990407
3. Kanaya, K. & Okayama, S. Penetration and energy-loss theory of electrons in solid targets. *J. Phys. D* 5, 43–58 (1972). doi:10.1088/0022-3727/5/1/308
4. Statham, P. J. Limitations to accuracy in extracting characteristic line intensities from X-ray spectra. *J. Res. NIST* 107, 531–546 (2002). doi:10.6028/jres.107.045
5. Goldstein, J. I. et al. *Scanning Electron Microscopy and X-Ray Microanalysis*, 4th ed. (Springer, 2018).
6. Newbury, D. E. & Ritchie, N. W. M. Performing elemental microanalysis with high accuracy and high precision by SEM/SDD-EDS. *J. Mater. Sci.* 50, 493–518 (2015). doi:10.1007/s10853-014-8685-2
7. Currie, L. A. Limits for qualitative detection and quantitative determination. *Anal. Chem.* 40, 586–593 (1968). doi:10.1021/ac60259a007